

The Limit Does Not Exist

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Dear Reader,

In the second quarter of 2026, Goldman Sachs Research noted that 68% of S&P 500 companies mentioned artificial intelligence (AI) on their calls. And AI capex spending now tops US\$1 trillion a year. It is fair to say that AI is the dominant force in the economy today. Against this backdrop, we wanted to share with you a bit more of how we think about AI.

Since the ChatGPT moment in November 2022, we've encountered plenty of traditional analysis which tells us the penetration of AI is still low, but also that companies investing in AI are generating a return on investment greater than their cost of capital. Of course, pinpointing these numbers and their magnitude is important for us as investors.

The other phenomenon we've observed is a tendency simply to speak loudly, confidently and in hyperbole about the upcoming and sweeping change. However, statements like "AI will change everything we have ever known" offer a conclusion without an explanation. Additionally, the reactions to such unmoored statements are predictable-fear, scoffing, indifference (as it's just too big and too unsubstantiated), and, for many contrarian investors, a deep desire to take the other side of that trade.

This paper seeks to avoid all of that.

Instead, it tries to explain the why: Why we believe AI will continue to improve, why it has a long runway of growth ahead and why it will make such an impact. As innovation investors, we seek to find new ideas that are 10x better than what came before. We think 10x is a useful shorthand for describing a fantastic new technology that will change people's behavior.

This paper explains why we have confidence not only that AI development will continue, but that it will improve 10x, not just once but every few years. We walk through why this repeated 10x improvement is unprecedented and powerful.

The rate of improvement of AI today is accelerating much faster than Moore's Law.¹ Unfortunately, laying out that framework takes more than a sentence and even more than a page, but we do think it is worth it.

This paper has two goals.

First, we want readers to understand that AI's continued exponential improvement is grounded in clear advances in compute infrastructure that, based on our research, are highly likely to continue. By the end of the paper, we think you will agree that, at a minimum, the improvements can continue at a spectacular rate for some period of time.

Second, we seek to offer a glimpse into the work and thinking of our team. We strive to be original, clear, and logically rigorous.

Above all, we hope this provides a stronger foundation for understanding the world we live in.

Finally, thank you to our team members Rich Ma and Tyler Whitehead for their contributions to this research.

Sincerely,

Matt Moberg Portfolio Manager Franklin Equity

The Why of AI (Parts I-II)

Why is this moment different? This paper explores why most technologies eventually plateau, why semiconductors have been the exception, and how the shift from Moore's Law to AI scaling may be helping usher in the Fourth Industrial Revolution.

The AI Scaling Era and the Advancements Ahead (Part III)

Where does the next wave of compute come from? This companion paper looks beyond the individual chip to the technologies reshaping the entire computing system and the multiple S-curves that could drive the next era of AI scaling.

Follow Matthew J. Moberg

Endnotes

Moore's Law: First articulated by Gordon Moore in 1965, it refers to the observation that the number of transistors on an integrated circuit doubles roughly every two years.

WHAT ARE THE RISKS?

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The allocation of assets among different strategies, asset classes and investments may not prove beneficial or produce desired results.

Equity securities are subject to price fluctuation and possible loss of principal. Investments in equity-linked notes often have risks similar to their underlying securities, which could include management risk, market risk and, as applicable, foreign securities and currency risks.

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